Michael Suguna Kumar V

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.83
Madras Christian College Higher Secondary School	Class XII (Tamil Nadu State Board (HSC))	2018	95.25%
Loyola Matriculation Higher Secondary School	Class X (Tamil Nadu State Board (SSLC))	2016	97.80%

WORK EXPERIENCE

Research Assistant, NeWS Lab, IIT Hyderabad

- Designed and deployed a **distributed 5G network testbed**, integrating real-time monitoring and telemetry systems across 8 network nodes.
- Built an ML-based **predictive fault detection pipeline** using **PCA, LSTM**, and **Random Forest**, achieving **98.73% accuracy / 98.71% F1** and predicting failures **5 seconds ahead**.
- Developed a **topology-agnostic anomaly detection system** that generalized across previously unseen network configurations, achieving **0.969 mean F1** without training separate models per configuration.
- Built a **domain-grounded RAG pipeline** over technical specifications, improving locally deployed **Small Language Model** accuracy from **60% --> 86%** on domain-specific benchmarks.

Backend Developer, Ramco Systems

- Re-architected payroll processing from **MS SQL stored procedures to a Java-based, multithreaded backend**, reducing processing time by **~94% (8 hours -> 30 minutes)** and improving scalability and client configurability.
- Developed a **no-code payroll automation platform**, enabling enterprise clients to configure and automate payroll workflows with minimal engineering effort.

Full Stack Developer, Lynk Logistics

- Migrated a monolithic Warehouse Management System to **microservices**, significantly improving system availability and reducing downtime.
- Designed and developed **Go REST APIs** for warehouse operations, using **GORM and PostgreSQL** for data persistence.
- Built scalable, responsive **React + Redux** UI applications used by **1,000+ warehouse workers**.

PROJECTS

Agentic AI for Root Cause Analysis & Mitigation in 5G O-RAN Systems

- Designed an **Agentic AI system** for automated root cause analysis, orchestrating specialized tools for **metric/log summarization, topology extraction, causal graph construction, and historical fault analysis** through multi-step investigation.
- Extended the agent into a closed loop diagnose and recover system, integrating system state, configuration, and execution tools and grounding LLM reasoning in **3GPP/O-RAN specifications** through fine tuning for reliable fault diagnosis and remediation.

RDMA Congestion Control Security

- Implemented a **telemetry-poisoning attack** against **HPCC** congestion control in **ns-3 (C++)**, manipulating in-network telemetry at a malicious switch.
- Demonstrated a **11.8% reduction** in sender congestion window and **11.7% increase** in flow completion time under falsified telemetry.

Network Traffic Prediction

- Designed and benchmarked **LSTM, GRU, and Transformer** architectures for **5-minute** ahead multivariate

time-series forecasting, comparing sequence-modeling approaches to predict resource utilization, user load, throughput, and latency from 30-second telemetry.

AI-Assisted Emergency Department Triage Assistant

- Architected a **multimodal AI-assisted triage platform** with separate patient and clinician dashboards, supporting **voice and image inputs** for symptom and wound assessment.
- Engineered a **hybrid AI-deterministic decision pipeline** where AI models generate structured findings while a fail-safe rules engine determines priority.

PUBLICATIONS

Yaswanth Kumar LS, Somya Jain, Michael Suguna Kumar Victor, Abdulla Ovais, Suresh Kumar Amalapuram, Bheemarjuna Reddy Tamma, Siva Ram Murthy C, "FALCON++: A Semi-Supervised Anomaly Detection and Localization Framework for Resilient O-RAN Deployments", accepted at **IEEE ICC 2026 - Open RAN Workshop**

RELEVANT COURSES

Foundations of Machine Learning (CS5590), Deep Learning (EE6380), AI in Finance (AI4803), Advanced Data Structures and Algorithms (CS6013), Advanced Computer Networks (CS5060), Networked Wireless Systems (CS5070), Security in Intelligent Systems (CS5913)

POSITIONS OF RESPONSIBILITY

IEEE Student Branch Chairperson

Joint Secretary, IT Department

ACHIEVEMENTS

Secured 3rd place in the ML Hackathon organized by **Meltwater**

Values Champ for the month

IEEE Best Student Volunteer

Infosys Certified Software Programmer

SKILLS

Languages: Python, C++, Java, Go, JavaScript, SQL

Software Engineering: Data Structures & Algorithms, OOP, REST APIs, Microservices, Multithreading, Socket Programming

Systems & Tools: Linux, Git, Docker, ns-3, Wireshark, ONNX, Prometheus, InfluxDB, Ollama, Kafka, Redis